

Credit Card Portfolio & Customer Analytics

Tableau Dashboard Documentation

SQL Analysis → Tableau Dashboard → Business Insights

Purpose: Convert the completed PostgreSQL/SQL analysis into an interactive Tableau dashboard using the card-level CSV dataset. The dashboard should tell the business story rather than simply reproduce the 25 SQL questions.

1. Dashboard Objective

The dashboard is designed to understand portfolio size and health, customer and product behaviour, digital adoption, credit exposure, issuance trends, and expiry/renewal opportunities.

The source analysis contains business questions covering portfolio size, customers, card types, networks, exposure, active/inactive status, contactless and virtual adoption, customer concentration, issuance, and expiry.

2. Source & Data Grain

Source: cards_data CSV / flat card-level dataset.

Field	Role	Tableau Type
card_id	Unique card identifier	String
customer_id	Customer identifier	String
card_type	Gold / Silver / Classic / Platinum	String
card_network	Network	String
credit_limit	Credit exposure / limit	Number
card_status	Active / Blocked / Expired / Lost	String
contactless	Yes / No	String
card_mode	Physical / Virtual	String
issue_date	Card issue date	Date
expiry_date	Card expiry date	Date

Important grain rule: one row represents one card. Portfolio metrics are card-level; customer metrics must be calculated at customer level when required.

3. Dashboard Architecture

Build four dashboard pages:

- Page 1 — Executive Portfolio Overview
- Page 2 — Customer & Product Analytics
- Page 3 — Digital Adoption & Behaviour
- Page 4 — Portfolio Health & Renewal

4. Page 1 — Executive Portfolio Overview

Goal: Give a management-level view of portfolio scale, value and current health.

KPI cards

KPI	Calculation	LOD?
Total Cards	COUNTD([Card ID])	No LOD
Total Customers	COUNTD([Customer ID])	No LOD
Total Credit Exposure	SUM([Credit Limit])	No LOD
Average Credit Limit	AVG([Credit Limit])	No LOD
Active Card %	Active Cards / Total Cards	Calculated field
Contactless Adoption %	Contactless Cards / Total Cards	Calculated field

Calculated fields

Active Cards

SUM(IF LOWER([Card Status]) = 'active' THEN 1 ELSE 0 END)

Active Card %

[Active Cards] / COUNTD([Card ID])

Contactless Cards

SUM(IF LOWER([Contactless]) = 'yes' THEN 1 ELSE 0 END)

Contactless Adoption %

[Contactless Cards] / COUNTD([Card ID])

Recommended visuals

- Credit Exposure by Card Type — compare SUM(Credit Limit) and optionally AVG(Credit Limit).
- Card Network Distribution — COUNTD(Card ID) by Card Network.
- Portfolio Health — card count and/or exposure by Card Status.

5. Page 2 — Customer & Product Analytics

Goal: Understand customer-level exposure, multi-card behaviour, product mix and concentration.

LOD expressions — use only where customer grain is required

Cards per Customer

{ FIXED [Customer ID] : COUNTD([Card ID]) }

Use: Used for cards-per-customer distribution.

Customer Exposure

{ FIXED [Customer ID] : SUM([Credit Limit]) }

Use: Used for top customers, customer exposure ranking and concentration.

Visuals

- Cards per Customer — distribution of 1, 2, 3+ cards using the Cards per Customer LOD.
- Top 10 Customers by Exposure — Customer ID vs Customer Exposure, descending.
- Card Type Exposure — total exposure by card type.
- Exposure per Card — SUM(Credit Limit) / COUNT(Card ID) by Card Type.

- Card Type × Network — highlight table or bar comparison.
- Customers with Multiple Cards — filter Cards per Customer > 1.

6. Page 3 — Digital Adoption & Behaviour

Goal: Show how customers/cards are adopting contactless and virtual card capabilities.

Calculated fields

Virtual Cards

SUM(IF LOWER([Card Mode]) = 'virtual' THEN 1 ELSE 0 END)

Virtual Adoption %

[Virtual Cards] / COUNTD([Card ID])

Contactless Cards

SUM(IF LOWER([Contactless]) = 'yes' THEN 1 ELSE 0 END)

Contactless Adoption %

[Contactless Cards] / COUNTD([Card ID])

Time-series fields

Issue Year

YEAR([Issue Date])

Issue Month

DATETRUNC('month', [Issue Date])

Visuals

- Virtual Card Adoption by Year — line chart.
- Contactless Adoption by Card Type — bar chart.
- Virtual vs Physical Cards — composition or bar chart.
- Optional monthly adoption trend if the dataset has sufficient monthly coverage.

7. Page 4 — Portfolio Health & Renewal

Goal: Identify inactive exposure and understand upcoming expiry/renewal workload.

Calculated fields

Inactive Cards

SUM(IF LOWER([Card Status]) <> 'active' THEN 1 ELSE 0 END)

Inactive Exposure

SUM(IF LOWER([Card Status]) <> 'active' THEN [Credit Limit] ELSE 0 END)

Expiry Year

YEAR([Expiry Date])

Expiry Month

DATETRUNC('month', [Expiry Date])

Visuals

- Inactive Exposure by Status — Blocked, Lost and Expired.
- Cards Expiring by Year — COUNTD(Card ID) by Expiry Year.
- Expiring Credit Exposure by Year — SUM(Credit Limit) by Expiry Year.
- Monthly Expiry Trend — optional line chart if monthly granularity is useful.
- Renewal Opportunity — combine expiring card count and expiring exposure.

8. Time-Series Analysis Plan

Time-series analysis should be more than a single yearly chart. Use annual trends for the strategic view and monthly trends where the data supports them.

Metric	Time field	Measure	Chart
Cards Issued	Issue Year / Issue Month	COUNTD(Card ID)	Line
Issued Credit Exposure	Issue Year / Issue Month	SUM(Credit Limit)	Line
YoY Issuance Growth	Issue Year	Year-over-year %	Line / KPI
Virtual Adoption	Issue Year	Virtual Adoption %	Line
Cards Expiring	Expiry Year / Month	COUNTD(Card ID)	Line / Bar
Expiring Exposure	Expiry Year / Month	SUM(Credit Limit)	Line / Bar

Important: do not use LOD simply because the chart is a time series. Normal aggregation is correct for issue/expiry counts and exposure. LOD is introduced only when the question changes grain, such as customer-level exposure or cards per customer.

9. Dashboard Filters

- Card Type
- Card Network
- Card Status
- Card Mode
- Contactless
- Issue Year / Issue Date
- Expiry Year / Expiry Date

Keep filters consistent across dashboards where they make business sense. Avoid adding every field as a filter; the dashboard should remain focused.

10. Recommended Dashboard Story

1. Start with portfolio scale: cards, customers and total exposure.
2. Move to value concentration: card type and customer exposure.
3. Explain customer behaviour: single-card vs multi-card customers.
4. Show digital behaviour: contactless and virtual adoption.
5. Move to time: issuance and exposure trends.

6. Finish with risk/opportunity: inactive exposure and upcoming expiry exposure.

11. Business Questions Covered

- Portfolio size and unique customers
- Average credit limit
- Card type and network distribution
- Highest credit exposure by card type
- Active-card percentage
- Contactless and virtual adoption
- Highest average credit limit by network
- Highest inactive rate by card type
- Cards per customer / multiple-card customers
- Top 10 customers by exposure
- Top 1% exposure concentration
- Exposure per card by card type
- Year-over-year card issuance
- Virtual adoption over time
- Contactless adoption by card type
- Inactive credit exposure
- Customer exposure ranking
- Top card type within each network
- Above-average customer exposure
- Expiry and renewal exposure

12. Important Dataset Notes

- There is no Premium card category in the dataset; do not create a premium-card visual unless the data is changed.
- Inactive means non-active statuses: Blocked, Expired and Lost.
- The customer card-count analysis should inspect the actual distribution before defining segments.
- Percentages should remain numeric in Tableau so they can be sorted, filtered and reused.
- LOD expressions should be used selectively for customer-grain calculations, not for every visual.

13. Final Build Order

1. Connect the CSV and validate field data types.
2. Create folders for KPI, Customer, Digital, Status and Time calculations.
3. Build Page 1 KPI cards and portfolio visuals.
4. Build the first LOD: Cards per Customer.
5. Build the second LOD: Customer Exposure.
6. Build customer/product visuals.

7. Build digital adoption visuals.
8. Build annual and monthly time-series visuals.
9. Build expiry and inactive-exposure visuals.
10. Add only the most useful filters.
11. Apply consistent formatting and titles.
12. Validate Tableau results against the completed SQL outputs before publishing.

14. Validation Checklist

- ☐ Total cards matches the SQL result.
- ☐ Unique customers matches the SQL result.
- ☐ Total credit exposure matches the SQL result.
- ☐ Average credit limit matches the SQL result.
- ☐ Active-card percentage matches the SQL result.
- ☐ Contactless adoption matches the SQL result.
- ☐ Virtual adoption matches the SQL result.
- ☐ Customer-level LOD results reconcile to SQL customer-level analysis.
- ☐ Annual issuance and YoY values reconcile to the SQL LAG() analysis.
- ☐ Expiry counts and exposure reconcile to the SQL expiry analysis.
- ☐ Currency is consistently displayed as ₹ throughout the dashboard.

Dashboard positioning: Credit Card Portfolio & Customer Analytics — SQL + Tableau + Business Insights